

PSC-Q: Public Service Calendars and Slack-Harvesting Substitution Padding for Congestion-Private Anonymous Overlays

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Abstract

We study practical mechanisms for meeting Congestion-EQ-style budgets in anonymous overlays, with emphasis on Anonymous-DHT lookups. A central engineering pitfall is that *additive* padding can worsen congestion leakage by pushing the system toward high utilization. We develop *PSC-Q* (Public Service Calendar for Queues): a mechanism family that (i) publishes a service calendar, (ii) harvests slack via *substitution* (idle-filling) rather than additive traffic, and (iii) uses declared dither to contract TV and to attenuate periodic witnesses. We connect PSC-Q to a knob-synthesis view (including slack-aware expiry synthesis, SAES) and provide new evidence quantifying the discretization “ δ -tax” for periodic calendars.

Compilation note. This paper is brief-by-design. A companion addendum collects the operator-facing microstudies and checklists that were previously shipped as a split supplement.[6]

Scope. We develop practical compiler mechanisms to meet Congestion-EQ budgets without shrinking slack: public service calendars and slack-harvesting *substitution* padding.

Imports. We import the Congestion-EQ contract (Paper I) as the target guarantee.[1] We also import the series’ destination/stop-time compiler interface and TV certificate vocabulary.[2, 3]

Exports. This paper exports PSC-Q mechanism families, slack-aware expiry synthesis (SAES), and calibration evidence for discretization/periodicity taxes when calendars are public.

Artifact policy. This archive is source-only; supporting plots and scripts are available on request.[35]

1 Introduction

PSC-Q is the mechanisms companion to the Congestion-EQ contract [1]. Where Congestion-EQ defines *what* must be bounded (a congestion-witness family plus an explicit accountant for repeats/resets), PSC-Q focuses on *how a compiler can actually meet those budgets* in a deployed overlay or Anonymous-DHT lookup service.

Figure omitted in source-only release. Requested file: `fig_additive_vs_substitution_tv.pdf`.

Figure 1: Archive evidence: substitution padding can reduce TV while additive padding can increase it at the same nominal “cover rate.”

Series map. We assume the series’ lookup-compiler interface for destination/contact traces and stop times [2, 3]. Congestion-EQ [1] introduces congestion as a first-class leakage tier. This paper proposes public-service-calendar mechanisms and slack-harvesting *substitution* padding to keep utilization in a safe regime. The assurance companion [5] shows how to publish certificate-ready knob choices, monitoring receipts, and transparency anchoring.

This paper is intentionally mechanisms-first. For motivation, threat surface, and the broader congestion/latency side-channel literature, see Congestion-EQ [1]. Here we focus on the compiler-facing issue: *padding that increases offered load can worsen congestion leakage* by shrinking slack. PSC-Q therefore emphasizes *substitution* (idle-filling) semantics and explicit slack targets; see also the companion assurance note W-Congestion-EQ [5].

Anonymous-DHT burstiness (pointer). Anonymous-DHT lookups can be bursty because they combine bounded parallelism, retries, and provider discovery. Rather than restate deployed-DHT measurement context, we point to Congestion-EQ §1 for the Anonymous-DHT motivation and citations, and to the Boss Fight dial-sheet paper for knob-level compilation context.

2 Padding taxonomy: additive vs substitution

Definition 1 (Additive vs substitution padding). *Additive* padding injects extra work on top of real work. *Substitution* padding (idle-filling) uses otherwise-idle capacity to serve cover traffic in place of idle time, without increasing mean load.

Remark 1 (Substitution must not steal slack). “Substitution” is slack-harvesting only if cover work is strictly limited to otherwise-idle service opportunities (e.g., calendar slots that would have been idle). If cover traffic competes with real work in the same queue (or contends for shared downstream resources), it can bias service and increase delay, defeating the congestion-privacy goal. Section 8 includes a small service-bias microstudy and an implementation checklist for “preemptible” cover.

The substitution/idle-filling idea mirrors classic constant-time countermeasures in software: when a secret-dependent branch would “finish early,” the implementation executes dummy work to fill the slack and equalize an adversary-visible timing trace [25, 26].

Substitution-style padding is deployed in Tor as hop-local connection- and circuit-level padding machines (negotiation in Proposal 254 and the Tor padding specification) [9, 10]. By contrast, classic website-fingerprinting defenses such as BuFLO and CS-BuFLO primarily add work to approach a constant-rate trace [11, 12]. This taxonomy matters for congestion privacy: additive padding can erode slack and worsen Congestion-EQ witnesses even when the padding is secret-independent.

The archive contains direct evidence that additive padding can increase distinguishability by shrinking slack.

3 PSC-Q: Public service calendars

PSC-Q fixes a public calendar of service opportunities (cells, quanta, or send slots) and then uses substitution semantics to fill idle opportunities with cover traffic. The calendar itself is a public

Figure omitted in source-only release. Requested file: `fig_tandem_padding.bottleneck_tv.pdf`.

Figure 2: Archive evidence: tandem/multi-hop systems can shift leakage to a different hop under naive padding. Mechanisms should track a witness family, not a single queue.

artifact and can be audited.

3.1 Why public calendars help

A public calendar separates two concerns: (i) schedule regularity and capacity provisioning are public and auditable; (ii) secret-dependent variability is forced into controlled degrees of freedom (e.g., which cover items occupy idle slots), enabling clean accounting.

3.2 Idle filling and partial-idle semantics

A PSC-Q calendar can be instantiated as a stream of public service opportunities (continuous-time rate μ or periodic ticks of width δ). At each opportunity, the relay serves one real cell if available; otherwise it emits a *hop-local* dummy that terminates at the next hop. If *every* idle opportunity is filled, the link-level departure timestamps match the public calendar exactly, eliminating hop-local rate/count leakage. Crucially, because dummies consume only idle service, the real-work queue dynamics are unchanged: this mechanism helps the *trace tier* without automatically fixing the *congestion tier* (Congestion-EQ).

Many deployments intentionally fill only a fraction of idle opportunities to save bandwidth. Partial idle filling reduces but does not eliminate departure-count distinguishability; Section 8 summarizes this residual leak and the resulting overhead tradeoffs.

Lemma 1 (Slack-harvesting semantics do not change the real queue). *Consider a single-server queue fed by real arrivals, with a public PSC-Q service calendar specifying service opportunities. At each opportunity the server serves one real job if available; otherwise it serves a hop-local dummy that consumes only that opportunity. Then (i) the real-queue length process and real waiting-time distribution are identical to the baseline system that simply idles when the real queue is empty, and (ii) the departure timestamps of served slots match the public calendar.*

Proof. Couple the PSC-Q system to the baseline on the same real-arrival sample path and the same calendar. Both systems serve a real job whenever one is present at a service opportunity; PSC-Q differs only by emitting a dummy when the real queue is empty. Thus the two systems have identical real departures and queue evolution, while PSC-Q replaces baseline idle slots with hop-local dummy service. $\square$

4 Multi-hop effects: bottleneck shift and localization

Even if hop-local congestion is controlled, multi-hop overlays can exhibit bottleneck shift: a defense that calms one queue can move congestion elsewhere.

5 Knob synthesis: slack-aware expiry synthesis (SAES)

PSC-Q exposes knobs: calendar period δ , flush strength, cover pool size, dither distribution, and expiry rules. The archive develops SAES as a helper that chooses expiry/timeout parameters to respect congestion budgets given slack.

Figure omitted in source-only release. Requested file: `fig_slack_aware_expiry_opt_tv.pdf`.

Figure 3: Archive evidence: SAES trades expiry aggressiveness against congestion distinguishability.

Figure omitted in source-only release. Requested file: `fig_dither_calendar_tv.pdf`.

Figure 4: Archive evidence: dithered calendars can contract congestion TV compared to rigid calendars.

6 Declared dither and the δ -tax for periodic calendars

A periodic service calendar introduces discretization artifacts: witnesses can exploit periodicity and quantization. Declared dither (publicly specified jitter) can reduce TV and attenuate spectral witnesses.

6.1 Declared dither as a contractive channel

A declared-dither mechanism can be viewed as a Markov kernel K that maps an “undithered” calendar-aligned quantity (or witness) X to an observed quantity Y . Total variation contracts under such channels.

Lemma 2 (TV contraction via the Dobrushin coefficient). *Let $K(\cdot | x)$ be a Markov kernel and define its Dobrushin (TV) contraction coefficient*

$$\eta_{\text{TV}}(K) := \sup_{x, x'} \text{TV}(K(\cdot | x), K(\cdot | x')) \in [0, 1].$$

Then for any two input laws P, Q on X ,

$$\text{TV}(PK, QK) \leq \eta_{\text{TV}}(K) \text{TV}(P, Q) \leq \text{TV}(P, Q).$$

If the same declared-dither kernel is applied m times (e.g., across ticks or hop-local stages), then $\text{TV}(PK^m, QK^m) \leq \eta_{\text{TV}}(K)^m \text{TV}(P, Q)$.

Proof. Standard; see Polyanskiy–Wu for SDPI and contraction coefficients [31]. The TV contraction coefficient is also closely related to Dobrushin’s ergodicity coefficient for Markov kernels [32, 33]. $\square$

Remark 2 (When does dither help?). The coefficient $\eta_{\text{TV}}(K)$ is strictly less than 1 only when different calendar states induce output distributions with nontrivial overlap at the observer’s effective resolution. This is why PSC-Q couples (δ, α) to an observer model and why monitoring must confirm that the deployment remained in the assumed timestamp-noise/jitter regime.

Remark 3 (Contraction coefficients as “privacy knobs”). Recent privacy work uses Dobrushin/TV contraction coefficients as computable summaries of how much a randomized mechanism collapses distinguishability, beyond the local-DP setting [34]. In PSC-Q, this motivates exporting $\eta_{\text{TV}}(K)$ (or a conservative upper bound) as a certificate-ready knob alongside (δ, α) .

6.2 Robustness to utilization drift

In deployment, utilization is not constant. Since the TV tier is slack-sensitive, a PSC-Q knob choice that is safe at one utilization can fail when background load increases. We therefore add a small robustness sweep varying ρ while holding (δ, α) fixed. Fig. 6 shows that TV grows rapidly as ρ approaches 1, and that dither can help but does not eliminate the need to monitor and re-tune.

Figure omitted in source-only release. Requested file: `fig_periodic_calendar_delta_tax.pdf`.

Figure 5: New evidence (added in this split archive): quantifying the discretization “ δ -tax” for periodic calendars. At fixed utilization $\rho = 0.8$ and a 1% arrival-rate shift, larger δ increases waiting-time distinguishability, while modest public dither ($\alpha = 0.25$) reduces it.

Figure omitted in source-only release. Requested file: `fig_delta_tax_utilization_drift.pdf`.

Figure 6: New evidence (added in this update): utilization-drift robustness for the TV-tier δ -tax experiment. For several representative (δ, α) settings, we plot waiting-time TV under a 1% arrival-rate shift as utilization ρ varies. This motivates compiling PSC-Q knobs against a utilization range and publishing *monitoring evidence* that the deployed regime stayed in-range.

7 Supplementary knob compilation results (integrated)

To avoid losing useful engineering-facing content during the transition to a unified archive, we inline the most deployable “supplement” material here in a compact form.

7.1 Representative knob table

Table 1 summarizes one simple PSC-Q knob sweep used by the archive evidence: for several timestamp-noise levels σ and utilization ceilings $\rho_{\max}$, it lists a feasible calendar family and key parameters.

σ	$\rho_{\max}$	family	δ	α	β	$1/\delta$
0.02	0.80	renewal	2.00	—	0.49	0.500
0.02	0.90	—	—	—	—	—
0.02	0.95	—	—	—	—	—
0.05	0.80	renewal	2.00	—	0.49	0.500
0.05	0.90	—	—	—	—	—
0.05	0.95	—	—	—	—	—
0.10	0.80	renewal	2.00	—	0.49	0.500
0.10	0.90	—	—	—	—	—
0.10	0.95	—	—	—	—	—

Table 1: Representative PSC-Q knob sweep (from the original split-paper evidence).

7.2 Knob manifest snippets and presets

PSC-Q can be deployed with a public “knob manifest” committing to the chosen calendar family, dither, and monitoring expectations. The manifest is a certificate-friendly object consumed by the assurance companion (W-Congestion-EQ).

Change control for calendar presets. Because knob manifests and calendar families are public artifacts, updates should run under the same disagreement-gated change-control discipline used for certified routing artifacts: log a proposed preset update, attach monitoring residuals (e.g., drift in slot-index recovery rates or utilization), and accept only if the disagreement gate is satisfied and the updated manifest is transparency-anchored [7, 8].

```

\begin{quote}\small
\texttt{version}: \texttt{pscq-knobs-v0.1}\\
\texttt{calendar.family}: \texttt{renewal\_uniform}\\
\texttt{calendar.delta\_ms}: \texttt{2}\\
\texttt{calendar.beta\_frac}: \texttt{0.49}\\
\texttt{utilization}: [0.7, 0.8]\\
\texttt{budgets}: (eps\_TV=0.05, eps\_spectral=0.15)\\
\texttt{knob\_table\_sha256}: \texttt{90a6a2c886069871...}\\
\end{quote}

```

```

\begin{quote}\small
\texttt{version}: \texttt{pscq-knobs-v0.1}\\
\texttt{calendar.family}: \texttt{renewal\_uniform}\\
\texttt{calendar.delta\_ms}: \texttt{2}\\
\texttt{calendar.beta\_frac}: \texttt{0.49}\\
\texttt{utilization}: [0.7, 0.8]\\
\texttt{budgets}: (eps\_TV=0.05, eps\_spectral=0.15)\\
\texttt{knob\_table\_sha256}: \texttt{90a6a2c886069871...}\\
\end{quote}

```

Other presets (e.g., higher observation noise or tighter utilization ceilings) use the same schema with different (δ, α, β) and different `knob_table_sha256` values. They are omitted here to keep the paper tight; the full preset table and reproduction scripts are available in the on-request artifact bundle.

7.3 Two short notes: utilization-robust compilation and slot-index recovery

Interval-robust compilation against utilization drift. A practical compiler target is a utilization range $\rho \in [\rho_{\min}, \rho_{\max}]$ rather than a point value. A minimal robustification is: pick (δ, α) to satisfy the Congestion-EQ TV contract at $\rho_{\max}$, and publish $\rho_{\max}$ as part of the manifest so that monitoring can trigger recertification when the regime changes [7]. This is the “range compilation” view used by the utilization-drift sweep in Fig. 6.

De-jittering and slot-index recovery (observer model). Let the calendar tick width be δ and let declared dither add uniform jitter in $[-\alpha\delta, \alpha\delta]$. If the observer has effective timestamp noise σ_{obs} (including instrumentation error and network jitter), then a crude sufficient condition for frequent slot-index ambiguity is

$$(1/2 - \alpha)\delta \lesssim \sigma_{\text{obs}},$$

since adjacent ticks overlap at the observer’s resolution. Conversely, if $\sigma_{\text{obs}} \ll (1/2 - \alpha)\delta$, an observer can often “snap” timestamps back to the nearest tick and recover slot indices, reintroducing periodic witnesses. This motivates treating (δ, α) as certificate-visible knobs and monitoring for drift in σ_{obs} .

8 Microstudies summary (plots on request)

The core PSC-Q argument is mechanism-level; nevertheless, a few deployment-facing microstudies are worth stating explicitly because they recur across systems. Figures and raw tables are omitted in this source-only archive but are available on request.

- **Additive padding can worsen congestion witnesses:** when additive cover pushes utilization into the high-delay knee, small secret-correlated load differences can create large shifts in waiting-time/RTT laws (Congestion-EQ tier). PSC-Q therefore prefers slack-harvesting substitution.
- **Partial idle-filling is often enough:** filling only a fraction of idle calendar slots can still materially reduce trace-tier distinguishability while preserving slack; the right fraction depends on the congestion budget and the downstream resource model.
- **Timestamp resolution matters:** if observers can recover slot indices from high-resolution timestamps, periodic/spectral structure returns. Declared dither should be sized so that slot ambiguity dominates effective observer timestamp noise.

9 Practical guidance for Anonymous-DHT compilers

- Prefer substitution/idle-filling to additive padding when the goal is congestion privacy.
- Provision explicit slack targets; do not treat “cover rate” as an independent knob.
- Treat δ and dither as first-class compiler outputs and include them in public receipts/certificates.
- If a periodic/spectral tier is in scope, choose α and δ so that slot ambiguity exceeds timestamp resolution. A crude rule-of-thumb from the slot-index recovery note above is to target $(1/2 - \alpha) \delta \lesssim \sigma_{\text{obs}}$, where σ_{obs} is the observer’s effective timestamp noise/jitter.
- Evaluate a witness family across hops (RTT time series, queue delay, loss/ECN) to avoid bottleneck shift.

10 Takeaways

- **Substitution beats additive for congestion privacy:** use idle-filling to reduce trace witnesses without shrinking slack.
- **Calendars are public interfaces:** when service opportunities are public and auditable, the remaining secrecy surface is pushed into controlled degrees of freedom that can be budgeted.
- **Dither is a first-class knob:** declared dither can contract TV-tier witnesses, but periodic/spectral tiers require explicit attention to timestamp resolution and slot recovery.
- **Treat utilization as part of the privacy contract:** congestion witnesses are highly nonlinear near the knee; publish slack targets and monitor drift.

11 Limitations and open questions

Declared dither contracts the *TV tier* for waiting-time witnesses, but it does not automatically protect against all witnesses. Periodic structure is routinely detectable in encrypted-traffic time series using autocorrelation and Lomb–Scargle periodograms [20, 21, 22, 23, 24], so a spectral tier is a realistic concern for overlays whose calendars induce periodicity. For a formal spectral-tier treatment in the series (including optimal distinguishing bounds under delegation), see [4]. As the slot-index recovery note above shows, an observer with high-resolution timestamps can often de-jitter and recover slot indices, regaining periodic structure. A full treatment therefore benefits

from a tiered view: (i) TV-tier accounting (contract paper), plus (ii) explicit periodic/spectral-tier goals, and (iii) monitoring for drift and unexpected witnesses (assurance paper).

12 Artifacts and reproducibility

This source-only release is intentionally small: it publishes the referee-facing arguments as a single LaTeX

file. Figures and full simulation artifacts used to generate the evidence are available on request as an external archive. To keep claims auditable, W-Congestion-EQ specifies the receipt fields a deployment should publish (knob manifest + monitoring summaries).

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